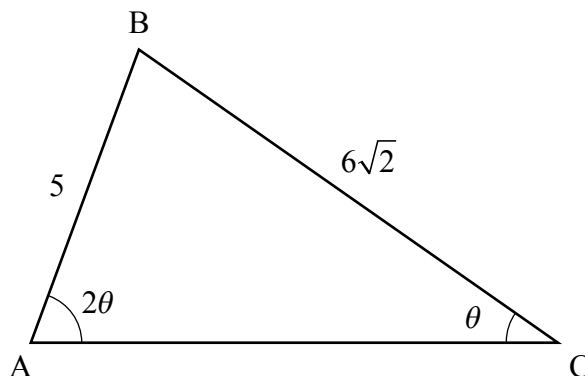


3. [Maximum mark: 7]

The following diagram shows a non-right angled triangle ABC .

diagram not to scale



$AB = 5$, $BC = 6\sqrt{2}$, $\hat{ACB} = \theta$ and $\hat{BAC} = 2\theta$, where $0 < \theta < \frac{\pi}{2}$.

(a) Using the sine rule, show that $\cos \theta = \frac{3\sqrt{2}}{5}$. [3]

(b) Hence, find $\sin \theta$. [2]

Point D is located on $[AC]$ such that the area of triangle BCD is $2\sqrt{14}$.

(c) Find DC. [2]

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